

Pre-Bootcamp Curriculum

Introduction

This curriculum provides essential preparation for joining the *develeap* DevOps Bootcamp. All resources listed are free and carefully selected to build the foundational skills required for success. While you are not required to complete every resource, familiarity with most content is strongly recommended.

Get to Know *develeap*

We recommend reviewing these resources to understand our bootcamp and company culture:

- [Bootcamp Website](#) - Comprehensive information and Q&A about the bootcamp
- [develeap Podcast - Bootcamp Episode](#) - Insights from our training approach
- [Magazine Articles](#) - Learn what to expect from our training
- [DevOps for Non-DevOps Podcast](#) - Familiarize yourself with core concepts and terminology
- **What is DevOps with Omri Spector:** [Part 1](#), [Part 2](#), [Part 3](#) - Our founder explains DevOps fundamentals through clear analogies and practical examples

Learning in the AI Era

AI tools like ChatGPT, Claude, and Gemini can accelerate your learning, but they work best when you understand the fundamentals. Use AI as a learning companion to explain concepts, debug code, and generate practice problems — but ensure you understand the solutions rather than just copying them. The bootcamp assessment process includes discussing your code and implementation choices, so genuine understanding is essential.

We strongly recommend getting familiar with [NotebookLM](#) — Google's AI-powered learning tool that can summarize documents, answer questions about your study materials, and even generate audio and video overviews to help you absorb complex topics.

[How Real DevOps Skills Are \(Actually\) Built](#) - Magazine Article

Phase 1: Bootcamp Admission Requirements

To gain admission to the bootcamp, you must successfully complete three stages:

1. **Scripting Assessment** - 2-hour programming test (preferably Python)
2. **Bash Scripting Task** - Home assignment demonstrating your Linux and Bash skills
3. **Technical Presentation Interview** - Present an assigned technical topic and participate in a comprehensive interview with a DevOps engineer

Core Programming Foundation

Scripting Languages

DevOps engineers primarily use Python and Bash for automation and system administration.

Python Programming Resources

- [CS50's Introduction to Python](#) - Harvard's comprehensive Python course

Bash Scripting Resources

- [The Linux Command Line](#) by William Shotts - Comprehensive Bash scripting reference
- [OverTheWire: Bandit](#) - Learn scripting through security challenges

Additional Practice:

- [develeap YouTube Playlist](#) - Video tutorials covering Bash, Python, and some Docker
- [Codewars](#) - Gamified coding challenges across multiple languages

Phase 2: What You'll Learn in the Bootcamp

Here's what awaits you once you successfully complete the admission process:

Bootcamp Syllabus

- **Bash**
Write effective scripts, handle text processing, and manage errors.
- **VCS & Git**
Setup, manage and use git - understanding basic branching strategies.
- **Docker**
Dockerize applications effectively.
- **Python**
Write python scripts, implement REST services, and function in a development team.
- **Networks, Application topologies, Docker Compose**
Implement popular runtime topologies using docker.
- **AWS Basics**
Configure secure networks and use central AWS services.
- **Jenkins, CI Pipelines**
Install, configure and use Jenkins to implement CI pipelines, Multi-branch pipelines, and advanced CI/CD patterns using groovy and cloud.
- **Infrastructure as Code**
Use Terraform to provision cloud resources and manage environments.
- **Kubernetes and Helm**
Setup and configure K8S clusters. Deploy stateless and stateful application topologies over the K8S cluster.
- **GitOps**
Implement the GitOps pattern as part of an overall CI/CD process involving Jenkins and ArgoCD.
- **Logging and Monitoring**
Setup and configure monitoring and alerting using ELK, Prometheus, and Grafana.

- **Certification**

Become certified with a highly-recognized and highly-valued certification, such as a Certified Kubernetes Administrator (CKA) or an AWS Solutions Architect Associate (SAA)

Technical Prerequisites

Operating System Setup

- **Windows Users:** Install Ubuntu dual-boot (virtual machines are insufficient and will cause performance issues)
- **Mac Users:** No Ubuntu installation required
- **M1 Mac Users:** Install Docker Desktop with maximum resource allocation to prevent image compatibility issues

Networking Fundamentals

Networking is integral to every DevOps tool and technology. Strengthen your foundation with these [OmniSecu](#) lessons:

- **Essential Lessons:** 1, 3, 7, 11, 12, 14-16, 18-20, 23, 25-27, 29, 32, 35-39, 52, 59, 61-64, 76, 83, 96, 101, 105, 107, 109, 110, 123

Version Control Fundamentals

Git is essential for all DevOps workflows and collaboration. Master these fundamentals before starting the bootcamp:

- [Git Official Documentation](#) - Comprehensive documentation and tutorials
- [Learn Git Branching](#) - Interactive visual tutorial focused on Git branching concepts

Advanced Preparation (Optional)

Docker Containerization

- [Docker Official Documentation](#) - Comprehensive introduction and best practices

AWS Cloud Fundamentals

- [AWS Cloud Quest - Cloud Practitioner](#) - Interactive learning through gamified scenarios

Support & Contact

For assistance throughout your preparation:

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